

APQ · AI POWERED QUALITY

Smart Glasses DVT Phase Gate Review

Comprehensive Quality Report

DEMONSTRATION CASE STUDY

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About This Document

Demonstration case study. This report was created for illustrative and educational purposes to showcase the APQ (AI Powered Quality) methodology — the fusion of quality-engineering expertise with AI-powered analysis. All data, metrics, failure rates, and costs are **hypothetical** and do not represent any real product or company. References to products or brands are illustrative only and imply no endorsement or affiliation. This document is provided "as is" and does not constitute professional engineering advice.

Executive Summary

GATE DECISION: CONDITIONAL PASS · OVERALL RISK: HIGH

Proceed to PVT only after all corrective actions are completed and validated against the entry criteria in Section 10.

DVT phase analysis reveals significant quality challenges requiring immediate attention before PVT transition. Audio distortion (12% failure rate), camera focus issues (5.2% failure), and battery degradation (15% below target) represent critical blockers. First Pass Yield of 78% falls substantially below the 95% target, with process capability (Cpk = 0.89) indicating inadequate process control. Root cause analysis identifies manufacturing process shortcuts — notably a reduction in adhesive cure time — and inadequate environmental testing as primary drivers.

Quality tools applied: 8D · 5-Why · FMEA · Pareto · SPC

Key Metrics Dashboard

Metric	Actual	Target	Status
Defect Density	1,200 DPMO	< 500 DPMO	Off track
First Pass Yield	78%	95%	Off track
Process Capability (Cpk)	0.89	> 1.33	Off track
Test Coverage	87%	> 95%	At risk

Critical Issues (open)	3	0	Off track
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Pareto Analysis

Audio quality (35% of defects) and camera reliability (22%) together account for **57% of all defects**. Immediate resources should focus on these two vital-few modes.

Defect Mode	Share	Cumulative	Relative scale	Priority
Audio distortion & speaker failures	35%	35%		Critical
Camera module defects	22%	57%		Critical
Battery performance issues	15%	72%		High
Hinge mechanism wear	12%	84%		Medium
Touch control issues	11%	95%		Medium

5-Why Root Cause Analysis

Problem statement: audio distortion in 12% of units at >70% volume.

Step	Question	Answer
Why 1	Why is there audio distortion?	Speaker diaphragm is delaminating from the voice coil.
Why 2	Why is delamination occurring?	Adhesive bond strength is insufficient.
Why 3	Why is bond strength insufficient?	Adhesive cure time was reduced from 4 hours to 2 hours.
Why 4	Why was cure time reduced?	To meet production takt time and schedule commitments.
Why 5	Why was engineering not consulted?	Production pressure prioritized schedule over process validation.

ROOT CAUSE

Manufacturing process shortcuts driven by schedule pressure — cure-time reduction made without engineering review or revalidation.

FMEA — Failure Mode & Effects Analysis

Failure Mode	S	O	D	RPN	Priority
Speaker diaphragm delamination	8	7	5	280	Critical
Battery capacity degradation	9	5	6	270	Critical
Touch control false triggers	5	7	5	175	High
Camera focus mechanism sticking	7	6	4	168	High
Hinge chrome plating wear	6	8	3	144	High

S = Severity (1–10) · O = Occurrence (1–10) · D = Detection (1–10) · RPN = S × O × D. Priority: RPN > 200 Critical · 100–200 High · < 100 Medium/Low.

8D Problem Solving

D1 — Team	Quality Engineering (lead), Manufacturing Engineering, Design Engineering, Supplier Quality, Product Management.
D2 — Problem	IS: audio distortion in 12% of units at >70% volume. IS NOT: not at lower volumes, not all units, not in EVT samples. Began Week 45.
D3 — Interim	100% audio testing at end-of-line; quarantine 347 suspected units; customer notification prepared; restore 4-hour cure time for work in progress.
D4 — Root cause	Adhesive cure time reduced from 4 hrs to 2 hrs → ~40% bond-strength reduction, verified through destructive pull testing.
D5 — Permanent	Restore 4-hour cure cycle with process control; install real-time temperature monitoring (± 2 °C); implement 100% bond pull testing (25 N minimum); formal ECN process for manufacturing changes.
D6 — Implement	Weeks 1–2: temperature monitoring (\$15K). Weeks 2–3: bond pull test stations (\$45K). Weeks 3–4: training and work instructions. Week 4: process validation (Cpk > 1.33).
D7 — Prevent	Manufacturing change-control policy; monthly process audits; quarterly supplier business reviews; share lessons learned across the organization.
D8 — Recognize	Quality achievement award at all-hands; root cause identified in 2 weeks (target: 4 weeks); team dinner scheduled.

Statistical Process Control

Station 3 — Speaker Assembly · Out of control

Eight consecutive points below the centerline indicate a systematic process shift. **Special cause:** adhesive cure-time reduction. Process capability Cpk = 0.72 — **inadequate**.

Station 7 — Camera Assembly · Marginally in control

Two points beyond the 2-sigma limits. **Common causes:** supplier lot variation, assembly fixture wear. Process capability Cpk = 1.15 — **marginal**.

Corrective Action Plan (CAPA)

ID	Action	Owner	Due	Budget
CA-1	Restore 4-hr cure cycle; temperature monitoring; bond pull testing	Mfg Eng	Week 51	\$60K
CA-2	Low-temperature lubricant; cold-chamber testing	Design Eng	Week 3	\$30K
CA-3	Battery charging circuit redesign; thermal management	Hardware Eng	Week 8	\$23K
CA-4	100% chrome plating thickness verification	Supplier QE	Week 50	\$35K
CA-5	SPC implementation at Stations 3 & 7	Mfg Eng	Week 51	\$35K
CA-6	Manufacturing change control; recurring audits	Quality Eng	Week 50	—

TOTAL INVESTMENT	PROJECTED ANNUAL WARRANTY SAVINGS	ROI
\$183K	\$4.8M	2,523%

Gate Decision — Entry Criteria for PVT

CONDITIONAL PASS

Timeline: 4 weeks to complete corrective actions and validation. Risk: **HIGH** if launched at current quality levels. Executive sponsorship required (schedule impact + \$183K capital). Weekly reviews mandatory until criteria are met.

- ☐ First Pass Yield > 90% demonstrated over 2 weeks of production (current: 78%)
- ☐ Process capability Cpk > 1.33 at all critical stations (current: 0.89)
- ☐ Zero critical open issues
- ☐ 100% of corrective actions implemented and validated

- ☐ Independent quality audit completed with score > 90%
- ☐ Supplier quality agreements updated and signed
- ☐ Environmental testing completed — 100% pass across -20°C to $+60^{\circ}\text{C}$

Lessons Learned

1. Manufacturing process changes — even under schedule pressure — must never bypass engineering approval and validation. Shortcuts cost more to fix than the time they save.
2. Environmental testing across the full operating range (-20°C to $+60^{\circ}\text{C}$, 0–95% RH) must be mandatory in EVT. Skipping it creates expensive field-failure risk.
3. Supplier quality needs proactive oversight. Supplier cost-reduction initiatives require customer engineering approval; sampling inspection (AQL) is inadequate for critical characteristics.
4. SPC and real-time monitoring are essential. Waiting for end-of-line test to catch process excursions produces scrap and rework — prevention beats detection.
5. Process capability ($\text{Cpk} > 1.33$) must be demonstrated before production release. Operating below $\text{Cpk} 1.0$ guarantees high defect rates and customer complaints.

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